



< Previous

✓

Next >

6.1.2 Overview

🔖 Bookmark this page

Pursue a verified certificate

✕

Verified Certificate

THE UNIVERSITY of TEXAS SYSTEM

Maurilio Pugliesi

Professor

UniversityX

Justine Doe, Ph.D.

Professor

UniversityX

This is to certify that

Sandipan Dey

successfully completed and received a passing grade in

UT.ALA:Advanced Linear Algebra: Foundations to Frontiers

a course of study offered by UTAustinX, an online learning initiative of University of Texas System.

edX

Verified Certificate

Valid Certificate ID

Issued Month / Year

12345678901234567890

➔ Unlimited access to all course materials

➔ Shareable [verified certificate](#)

➔ AI learning assistant and translations

Upgrade for \$99

- **6.1 Opening Remarks**

- 6.1.1 Whose problem is it anyway?
- 6.1.2 Overview
- 6.1.3 What you will learn

- **6.2 Floating Point Arithmetic**

- 6.2.1 Storing real numbers as floating point numbers
- 6.2.2 Error in storing a real number as a floating point number
- 6.2.3 Models of floating point computation
- 6.2.4 Stability of a numerical algorithm
- 6.2.5 Conditioning versus stability
- 6.2.6 Absolute value of vectors and matrices

- **6.3 Error Analysis for Basic Linear Algebra Algorithms**

- 6.3.1 Initial insights
- 6.3.2 Backward error analysis of dot product: general case
- 6.3.3 Dot product: error results
- 6.3.4 Matrix-vector multiplication
- 6.3.5 Matrix-matrix multiplication

- **6.4 Error Analysis for Solving Linear Systems**

- 6.4.1 Numerical stability of triangular solve
- 6.4.2 Numerical stability of LU factorization
- 6.4.3 Numerical stability of linear solve via LU factorization
- 6.4.4 Numerical stability of linear solve via LU factorization with partial pivoting
- 6.4.5 Is LU with Partial Pivoting Stable?

- **6.5 Enrichments**

- 6.5.1 Systematic derivation of backward error analyses
- 6.5.2 LU factorization with pivoting can fail in practice

- **6.6 Wrap Up**

- 6.6.1 Additional homework
 - 6.6.2 Summary
-

Discussion

Hide Discussion

Topic: Week 06 / 06.1.2

Add a Post

Show all posts ▼by recent activity ▼

There are no posts in this topic yet.

✕

< Previous

Next >

© All Rights Reserved



edX

- About
- Affiliates
- edX for Business
- Open edX
- Careers
- News

Legal

- Terms of Service & Honor Code
- Privacy Policy
- Accessibility Policy
- Trademark Policy
- Sitemap
- Cookie Policy
- Your Privacy Choices

Connect

- Idea Hub
- Contact Us
- Help Center
- Security
- Media Kit

